Machine Learning Overview

Machine Learning (ML) is a type of software that predicts outcomes based on data. The more experience (data) provided to the machine, the more accurate its predictions become. Essentially, the machine learns from past data to improve its performance over time.

Types of Learning in Machine Learning

1. **Supervised Learning:**
   In supervised learning, the model is trained on labeled data, where both the input and output are provided. The goal is to learn the relationship between input variables and the corresponding output.
2. **Unsupervised Learning:**
   This involves training a model on data that is not labeled. The model tries to find hidden patterns or relationships within the data.
3. **Semi-Supervised Learning:**
   Semi-supervised learning uses both labeled and unlabeled data. The model improves by leveraging the large amount of unlabeled data, while still relying on the smaller amount of labeled data.
4. **Reinforcement Learning:**
   In reinforcement learning, an agent performs actions in an environment and receives rewards or penalties based on its actions. The goal of the agent is to maximize its cumulative reward over time. This is achieved by learning from both positive and negative feedback, gradually improving its decision-making policy.

Types of Model Training Approaches

1. **Batch Learning:**
   In batch learning, the entire dataset is used to train the model at once. This approach requires loading all the data into memory, which can be problematic if the dataset is large (e.g., 1-2 TB). Additionally, once trained, the model cannot learn from new data without retraining on the combined old and new data.
2. **Online Learning:**
   In online learning, the model is trained incrementally as new data becomes available. This allows for continuous learning without needing to retrain the entire model. New data can be easily integrated, and memory is not as big of an issue as in batch learning.
   - **Disadvantages of Online Learning:**
     - It can be challenging to set the priority or learning rate for new data. If new data with errors is given high priority, the model may be negatively influenced. Conversely, if high-quality data is given low priority, the model may not learn effectively from it.

Model-Based vs Instance-Based Algorithms

1. **Model-Based Algorithms:**
   These algorithms use a predictive model to make predictions. The model doesn't require access to all past data once trained.
2. **Instance-Based Algorithms:**
   Instance-based algorithms rely on the entire dataset to make predictions. This approach can lead to memory issues since all data needs to be stored and used during the prediction process.

In summary, machine learning offers various learning methods and training techniques that can be tailored to different use cases, depending on factors like the size of the data, memory constraints, and the desired learning approach.

Fundamental Challenges in Machine Learning

In machine learning, several fundamental challenges can arise, both in terms of the data and the model. Addressing these issues is crucial for building accurate and reliable models.

Challenges with Data

1. **Bad Data:**
   Poor quality data can lead to incorrect predictions. Data needs to be cleaned and preprocessed effectively to ensure good results.
2. **Insufficient Data:**
   A lack of adequate data makes it difficult for the model to learn patterns, leading to underperformance.
3. **Non-Representative Data:**
   If the training data doesn’t accurately represent the real-world population or scenario, the model may not generalize well.
4. **Sampling Noise:**
   When too little data is sampled, there is a higher risk of including noise, which can distort the learning process and lead to inaccurate predictions.
5. **Sampling Bias:**
   Sampling bias occurs when a subset of data that does not represent the diversity of the entire dataset is used. This can result in skewed or biased models.
6. **Irrelevant Data:**
   Including irrelevant features in the training data can confuse the model and lower its performance. Feature selection is necessary to focus on important data points.

Challenges with the Model

1. **Overfitting:**
   Overfitting happens when a model is too complex and learns not only the underlying patterns but also the noise in the data. This leads to poor generalization on new data.
2. **Underfitting:**
   Underfitting occurs when the model is too simple to capture the underlying trends in the data, resulting in poor performance on both training and unseen data.
3. **Model Generalization:**
   To achieve a balance between overfitting and underfitting, a model must generalize well. This means it should perform well not just on the training data but also on unseen test data.
4. **Imbalanced Datasets:**
   Random sampling doesn't work well with imbalanced datasets, where one class is represented far more than others. In such cases, **stratified sampling** should be used to ensure that each class is proportionally represented.

Validation Data and Testing

1. **Hyperparameter Tuning:**
   Hyperparameters, which are set before training begins, need to be carefully tuned to optimize the model's performance.
2. **Model Selection:**
   Choosing the right model for the task is critical. The selection process involves comparing the performance of different models on validation data.
3. **Test Data:**
   A model should be tested on a separate set of data, known as test data, to assess how well it generalizes to unseen data after training and validation.

In summary, successful machine learning requires attention to both data quality and model complexity. Addressing these challenges ensures that the model is both accurate and generalizable across various data inputs.

Distance Metrics in Machine Learning

1. **Euclidean Distance:**
   This metric calculates the shortest straight-line distance between two points in Euclidean space. It is commonly used for continuous data in machine learning algorithms.
2. **Manhattan Distance:**
   Also known as "L1 distance," it calculates the distance between two points by summing the absolute differences of their coordinates. It measures the vertical and horizontal distances, like navigating a city grid.

Data Leakage

- **Data Leakage** occurs when information from the test dataset unintentionally influences the model during training or hyperparameter tuning.
- The **test dataset** should remain unseen by the model to accurately measure how well the model generalizes to new data.
- To avoid data leakage, the **validation dataset** is used to tune hyperparameters and select models. The test dataset should only be used for final evaluation.

Data Mismatch

- **Data Mismatch** happens when the test or validation data does not accurately represent the real-world scenario.
- For instance, using highly processed web images for training but testing on real-life images may lead to poor model performance.
- It is crucial to use **real-life** and **representative data** for testing to ensure the model can generalize properly to unseen scenarios.

Train-Dev Set

- The **Train-Dev set** is used to evaluate whether the model is overfitting.
- If the train-dev accuracy is much lower than the actual training set accuracy, it indicates **overfitting**. This means the model is learning too much from the training data and may not perform well on unseen data.

Model Evaluation Metrics

When evaluating a machine learning model, we often use various metrics to measure how well the model performs. Here are some key concepts and metrics:

1. **Error Magnitude:**
   When calculating errors, we are only interested in the magnitude, not whether the error is positive or negative. The absolute value of errors is considered to avoid canceling out errors (positive and negative errors balancing each other).
2. **Mean Absolute Error (MAE):**
   - This metric calculates the average of the absolute differences between the predicted and actual values.
   - MAE is helpful because it is easy to interpret, but it doesn't highlight larger errors as much as other metrics.
3. **Mean Squared Error (MSE):**
   - MSE is the average of the squared differences between predicted and actual values.5

    - It is the square of MAE, and by squaring the errors, MSE gives more weight to larger errors, making it easier to catch significant errors.
    - One downside is that it is harder to interpret in the original scale since the errors are squared.
4. **Root Mean Squared Error (RMSE):**
    - RMSE is the square root of MSE.
    - It brings the error back to the original scale of the data, making it more interpretable for reporting purposes.
    - RMSE is often used in reports because it provides a clearer sense of how much error can be expected in the original units.

Convex Graph & Parameter Tuning

- When tuning hyperparameters of a model, running the model with different parameter values will often create a **convex graph** between the **error** and the **parameter values**.
- The goal is to minimize the error, and the **global minima** (lowest point on the graph) represents the best parameter value that results in the least error.
- Identifying and using the **global minima** helps optimize the model performance.